

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo + Discrepancy

Improving Efficiency

Selected Topics

Git website and repository
Canvas

Fred Hickernell, Fall 2025

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Markov Chain Monte Carlo (MCMC) + Discrepancy

Owen, Chapters 11– 12

Project Selection due Oct 1

Assignment 3 due Oct 17



Rate your confidence heading into Test 1

- Go to [menti.com](https://www.menti.com)
- Use code 4923 2283



How do we generate samples when the distribution is complicated?



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 - An unnormalized target density
 - May reject a lot of samples



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- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling
 - Propose from a simple density that you know bounds
 - An unnormalized target density
 - May reject a lot of samples
- **Markov chain Monte Carlo** (MCMC) moves samples toward areas of higher target density



Markov Chains

Let $\mathcal{X}$ be a state space for a **sequence** of random variables. Markov chains follow a probability law satisfying

$$\mathbb{P}(X_{n+1} \in \mathcal{A} \mid X_n = \mathbf{x}_n, \dots, X_0 = \mathbf{x}_0) = \mathbb{P}(X_{n+1} \in \mathcal{A} \mid X_n = \mathbf{x}_n)$$

where $\mathcal{A}$ is some (measurable) subset of $\mathcal{X}$. This means that **where you go next only depends on where you are now**, not how you got there.

We use these to

- Generate $X_0, X_1, \dots$ whose **probability distribution** converges to the target
- Study **queuing** systems

We are often interested in the **asymptotic** behavior of the X_i



Metropolis–Hastings algorithm

Suppose that

- You really **want to** sample X with known (unnormalized) **target** density q
- You have a **proposal** density $q_{\text{new|old}}$ to select the next point starting from where you are

Given $X[0]$

For $i = 0$ to $n-1$

Generate $Z \sim q_{\text{new|old}}(\cdot | X_i)$, $U \sim \mathcal{U}[0,1]$

If $U \leq \min\left(1, \frac{q(Z)q_{\text{new|old}}(X_i | Z)}{q(X_i)q_{\text{new|old}}(Z | X_i)}\right) = \min\left(1, \frac{\text{joint density of first } Z \text{ then } X_i}{\text{joint density of first } X_i \text{ then } Z}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$

Note that you **always accept** either the new or the old



Metropolis algorithm

If $q_{\text{new}|\text{old}}(x | z) = q_{\text{new}|\text{old}}(z | x)$, then this can be simplified

Metropolis sampling proceeds as follows

Given $X[0]$

For $i = 0$ to $n-1$

Generate $Z \sim q_{\text{new}|\text{old}}(\cdot | X_i)$, $U \sim \mathcal{U}[0,1]$

If $U \leq \min\left(1, \frac{q(Z)}{q(X_i)}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$



Metropolis algorithm practice

Consider some uniform random numbers

i	0	1	2	3	4	5	6	7
U_i	0.6	0.1	0.3	0.5	0.9	0.2	0.1	0.4

and some values of a density

x	-1.2	-0.8	-0.5	0.6	1.1	1.4	2.0	2.2
$\varrho(x)$	0.1	0.8	2	1.6	0.5	2.6	2.4	1.0

Choose a random order of the x values and follow the Metropolis algorithm for a several steps based on the table of density values



Advantages of and Challenges in MCMC

- Samples from complicated, unnormalized densities
- Easy to implement
- Proposal density and starting point need tuning to ensure that
 - Explore all the sample space and not miss out
 - Exploit sampling in regions where the density is large



How do we measure quality of MCMC

Short answer: We cannot easily

Longer answer: Given a symmetric, positive definite kernel, $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, the **discrepancy**, sometimes called the maximum mean discrepancy between two empirical distributions is defined by

$$D^2(\{\mathbf{x}_i\}_{i=0}^{m-1}, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(\mathbf{x}_i, \mathbf{z}_j) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

Requires $\mathcal{O}(\max(m, n)^2)$ operations



What is discrepancy?

The **discrepancy** measures

- The **difference** between two probability distributions
- The **worst-case error** in approximating an expectation/integral $\mu = \mathbb{E}[f(X)]$ by a sample mean when f is in the unit ball of a Hilbert space with **reproducing kernel** K
- The **root-mean square error** in approximating an expectation/integral $\mu = \mathbb{E}[f(X)]$ by a sample mean when f is an instance of a Gaussian process with **covariance kernel** K

Discrepancies from kernels were popularized by [Hickernell 1998] and [Gretton et al. 2012]



Symmetric, Positive Definite Kernels

- A square matrix, K , is symmetric and positive definite iff

$$K^T = K \quad \text{and} \quad \mathbf{c}^T K \mathbf{c} > 0 \text{ for all } \mathbf{c} \neq \mathbf{0}$$

- A kernel, $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, is **symmetric and positive definite** iff

$$K := \left(K(\mathbf{x}_i, \mathbf{x}_j) \right)_{i,j=0}^{n-1}, \quad \text{the Gram matrix, is symmetric and positive definite}$$

for all $n \in \mathbb{N}$ and $\{\mathbf{x}_i\}_{i=0}^{n-1}$ with distinct elements, e.g.

$$K(\mathbf{t}, \mathbf{x}) = \exp\left(-\|\mathbf{t} - \mathbf{x}\|^2/h^2\right), \quad \mathcal{X} = \mathbb{R}^d, \quad \text{squared exponential}$$

$$K(\mathbf{t}, \mathbf{x}) = \prod_{\ell=1}^d \left[1 + \frac{\gamma_{\ell}^2}{2} \left(|t_{\ell} - 1/2| + |x_{\ell} - 1/2| - |t_{\ell} - x_{\ell}| \right) \right], \quad \mathcal{X} = [0,1]^d$$

centered discrepancy



Discrepancy from kernels

Suppose that

- $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a symmetric, positive definite kernel,
- $\mathcal{X}$ is the sample space for random variables/vectors $\mathbf{X}$ and $\mathbf{Z}$
- $\mathbf{X}$ has CDF F_X and $\mathbf{Z}$ has CDF F_Z

Then **discrepancy** measures the difference between two distributions:

$$\begin{aligned} D^2(F_X, F_Z; K) &= \mathbb{E}_{X, X' \sim F_X} K(X, X') - 2\mathbb{E}_{X \sim F_X, Z \sim F_Z} K(X, Z) + \mathbb{E}_{Z, Z' \sim F_Z} K(Z, Z') \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}') - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) dF_X(\mathbf{x}) dF_Z(\mathbf{z}) \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') dF_Z(\mathbf{z}) dF_Z(\mathbf{z}') \geq 0 \end{aligned}$$



Discrepancy from kernels

Then **discrepancy** measures the difference between two distributions:

$$\begin{aligned} D^2(F_X, F_Z; K) &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}') - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) dF_X(\mathbf{x}) dF_Z(\mathbf{z}) \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') dF_Z(\mathbf{z}) dF_Z(\mathbf{z}') \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') d[F_X(\mathbf{x}) - F_Z(\mathbf{x})] d[F_X(\mathbf{x}') - F_Z(\mathbf{x}')] \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \varrho_X(\mathbf{x}) \varrho_X(\mathbf{x}') d\mathbf{x} d\mathbf{x}' - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) \varrho_X(\mathbf{x}) \varrho_Z(\mathbf{z}) d\mathbf{x} d\mathbf{z} \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') \varrho_Z(\mathbf{z}) \varrho_Z(\mathbf{z}') d\mathbf{z} d\mathbf{z}' \geq 0 \end{aligned}$$



Discrepancy with empirical distributions

The formula for the squared discrepancy works if one or both of the distributions is the **empirical distribution** of a sample.

$$\begin{aligned} D^2(F_X, F_{\{z_i\}_{i=0}^{n-1}}; K) &= D^2(F_X, \{z_i\}_{i=0}^{n-1}; K) \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(x, x') dF_X(x) dF_X(x') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(x, z_i) dF_X(x) \\ &\quad + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(z_i, z_j) \end{aligned}$$

$$D^2(\{x_i\}_{i=0}^{m-1}, \{z_i\}_{i=0}^{n-1}; K) = \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(x_i, x_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(x_i, z_j) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(z_i, z_j)$$



Unbiased estimates of squared discrepancy

The squared discrepancy is non-negative, but we can construct **unbiased and possibly negative** estimates of $D^2(F_X, F_Z; K)$ based on IID samples $\{\mathbf{x}_i\}_{i=0}^{m-1}$ and/or $\{\mathbf{z}_i\}_{i=0}^{n-1}$

$$D_{\text{unb}}^2(F_X, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') d\varrho_X(\mathbf{x}) d\varrho_X(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) d\varrho_X(\mathbf{x}) \\ + \frac{1}{n(n-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

$$D_{\text{unb}}^2(\{\mathbf{x}_i\}_{i=0}^{m-1}, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \frac{1}{m(m-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{m-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(\mathbf{x}_i, \mathbf{z}_j) \\ + \frac{1}{n(n-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$



Discrepancy as worst case error

Let $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a **symmetric, positive definite** kernel. Then

- $\mathcal{H} :=$ **completion** (add limits of Cauchy sequences) of
$$\{c_0 K(\cdot, \mathbf{x}_0) + \cdots + c_{n-1} K(\cdot, \mathbf{x}_{n-1}) : n \in \mathbb{N}, c_i \in \mathbb{R}, \mathbf{x}_i \in \mathcal{X}\}$$
is a **Hilbert space** (vector space + inner product) of functions defined on $\mathcal{X}$ with **reproducing kernel** K
- Which means $f(\mathbf{x}) = \langle K(\cdot, \mathbf{x}), f \rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}, \mathbf{x} \in \mathcal{X}$
- Suppose that $\int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}')$ is defined and finite, which means that $f \mapsto \int_{\mathcal{X}} f(\mathbf{x}) dF_X(\mathbf{x})$ is a **bounded linear functional** on $\mathcal{H}$
- This means that **cubature error**, $f \mapsto \int_{\mathcal{X}} f(\mathbf{x}) dF_X(\mathbf{x}) - n^{-1} \sum_{i=0}^{n-1} f(\mathbf{z}_i)$ is also a bounded linear functional on $\mathcal{H}$



Discrepancy as worst case error (cont'd)

- **Reproducing property:** $f(\mathbf{x}) = \langle K(\cdot, \mathbf{x}), f \rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$, $\mathbf{x} \in \mathcal{X}$
- **Cubature error,** $f \mapsto \int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - n^{-1} \sum_{i=0}^{n-1} f(\mathbf{z}_i)$, is a bounded linear functional on $\mathcal{H}$
- By the **Riesz Representation Theorem** there exists a $\zeta \in \mathcal{H}$ such that

$$\int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) = \langle \zeta, f \rangle_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

- The **Cauchy-Schwarz inequality** separates the contribution to the error from $\{\mathbf{z}_i\}_{i=0}^{n-1}$ and from f

$$\left| \int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right| = \left| \langle \zeta, f \rangle_{\mathcal{H}} \right| \leq \|\zeta\|_{\mathcal{H}} \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$



Discrepancy as worst case error (cont'd)

$$\left| \int_{\mathcal{X}} f(\mathbf{x}) \, dF_X(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right| = \left| \langle \zeta, f \rangle_{\mathcal{H}} \right| \leq \|\zeta\|_{\mathcal{H}} \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

- By the reproducing property and the Riesz Representation Theorem,

$$\zeta(\mathbf{x}') = \langle K(\cdot, \mathbf{x}'), \zeta \rangle_{\mathcal{H}} = \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}') \, dF_X(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} K(\mathbf{z}_i, \mathbf{x}')$$

$$\begin{aligned} \|\zeta\|_{\mathcal{H}}^2 = \langle \zeta, \zeta \rangle_{\mathcal{H}} &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \, dF_X(\mathbf{x}) dF_X(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) \, dF_X(\mathbf{x}) \\ &\quad + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j) = D^2(F_X, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) \end{aligned}$$



Discrepancy as worst case error (cont'd)

In particular, for **uniform target** distributions

$$\left| \int_{[0,1]^d \times [0,1]^d} f(\mathbf{x}) \, d(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right| \leq D(\mathcal{U}[0,1]^d, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

where

$$\begin{aligned} D^2(\mathcal{U}[0,1]^d, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = & \int_{[0,1]^d \times [0,1]^d} K(\mathbf{x}, \mathbf{x}') \, d\mathbf{x} d\mathbf{x}' - \frac{2}{n} \sum_{i=0}^{n-1} \int_{[0,1]^d} K(\mathbf{x}, \mathbf{z}_i) \, d\mathbf{x} \\ & + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j) \end{aligned}$$



After Test 1, how do we improve?

- Go to menti.com
- Use code 8510 7177
- Two slides
 - One to help me know how I can help
 - One for you to reflect on what you will do differently



My response to your feedback

- You requested more worked examples, especially those that mimic test problems
 - You need to study to learn **concepts**, not to be able to do problems
 - I will do **more examples** in class; you are welcome to ask for more
 - I will assign more problems for homework of the **type** that may appear on tests (see the revised Assignment 3)
 - I encourage you to look at **old tests**
 - My tests **will not intentionally copy** class examples or old test problems



My response to your feedback

- You asked for more background material and were concerned that the class required advanced mathematics or statistics.
- The statistics and mathematics required on the tests are a calculus and a calculus-level probability class
 - How to take **expectations**
 - The binomial, exponential, and normal/Gaussian **distributions**
 - **Sample** means and variances
 - **Confidence** intervals via the Central Limit Theorem
- I do not expect you to know the details of functional analysis or stochastic processes, but to understand the overall argument
- To help me understand what advanced mathematics or statistics you need help with please respond at [mentimeter.com](https://www.mentimeter.com) 2644 2309



My response to your feedback

- There were comments about the test format, how to recover from a low score and number of tests
 - **In-person** tests help me gauge what you have learned
 - I will take the **highest** of your two test scores rather than average the two
 - This is a **graduate** class suitable for masters and doctoral students
- Please provide further and continuing feedback through
 - Comments in class
 - Emails
 - Future Mentimeter
 - Midterm course evaluation from the Center for Learning Innovation



Discrepancy as average case error

Suppose that

- f is drawn from a Gaussian process with zero mean and **covariance kernel** $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, i.e., $f \sim \mathcal{GP}(0, K)$.
- Do not need to specify the sample space for f , which will almost surely have less smoothness than the Hilbert space with reproducing kernel K .

The mean squared error for a deterministic cubature rule is

$$\begin{aligned} & \mathbb{E}_{f \in \mathcal{GP}(0, K)} \left| \int_{\mathcal{X}} f(\mathbf{x}) \, dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right|^2 \\ &= \int_{\mathcal{X} \times \mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{x}')] \, dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{z}_i)] \, dF_{\mathbf{X}}(\mathbf{x}) \\ & \quad + \frac{1}{n^2} \sum_{i, j=0}^{n-1} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{z}_i) f(\mathbf{z}_j)] \end{aligned}$$



Discrepancy as average case error (cont'd)

The mean squared error for a deterministic cubature rule is

$$\begin{aligned} & \mathbb{E}_{f \in \mathcal{GP}(0, K)} \left| \int_{\mathcal{X}} f(\mathbf{x}) \, dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right|^2 \\ &= \int_{\mathcal{X} \times \mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{x}')] \, dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{z}_i)] \, dF_{\mathbf{X}}(\mathbf{x}) \\ & \quad + \frac{1}{n^2} \sum_{i, j=0}^{n-1} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{z}_i) f(\mathbf{z}_j)] \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \, dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) \, dF_{\mathbf{X}}(\mathbf{x}) + \frac{1}{n^2} \sum_{i, j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j) \\ & \quad = D^2(F_{\mathbf{X}}, \{\mathbf{z}_i\}_{i=0}^{n-1}) \end{aligned}$$



Observations about discrepancy

- The value of the discrepancy depends on the **choice of kernel**, K , and the parameters that define it
 - K may be chosen for convenience
 - Should choose the **hyperparameters** of K to ensure that your f is in the **middle**; this can be done by **empirical Bayes/maximum likelihood estimation** in the average case setting
 - K may be chosen based on knowledge about **properties** of f , such as domain, smoothness, periodicity, importance of different coordinates
 - $D(\cdot, \cdot; c^2 K) = |c| D(\cdot, \cdot; K)$
- $D(F_X, \{z\}_{i=0}^{n-1}; K)$ **measures the quality** of $\{z\}_{i=0}^{n-1}$ for estimating the mean, $\mu = \mathbb{E}[f(X)]$, by the sample mean $\hat{\mu}_n = n^{-1} \sum_{i=0}^{n-1} f(z)$ for
 - f in a Hilbert space, $\mathcal{H}$, with reproducing kernel K (don't need to know $\|\cdot\|_{\mathcal{H}}$ explicitly) or
 - $f \sim \mathcal{GP}(0, K)$



Observations about discrepancy (cont'd)

- The mean square discrepancy of an **IID random** sample is

$$\mathbb{E}\left[D^2\left(F_{\mathbf{X}}, \{z_i \stackrel{\text{IID}}{\sim} F_{\mathbf{X}}\}_{i=0}^{n-1}; K\right)\right] = \frac{1}{n} \left[\int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') \right]$$

- The deterministic cubature error bound **cannot** be used constructively to bound the error because the norm of f cannot be estimated
- If $f \sim \mathcal{GP}(0, K)$, and the parameters of K are estimated by empirical Bayes (maximum likelihood), then one can construct **credible intervals** for the cubature error

$$\mathbb{P}_{f \in \mathcal{GP}(0, K)} \left[\left| \int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(z_i) \right| \leq \frac{2.58 D(F_{\mathbf{X}}, \{z_i\}_{i=0}^{n-1}; K)}{\sqrt{n}} \right] \geq 99 \%$$



Example of the Centered discrepancy

$$K(t, \mathbf{x}) = \prod_{\ell=1}^d \left[1 + \frac{\gamma_{\ell}^2}{2} (|t_{\ell} - 1/2| + |x_{\ell} - 1/2| - |t_{\ell} - x_{\ell}|) \right], \quad \mathcal{X} = [0, 1]^d$$

$$\|f\|_{\mathcal{H}}^2 = \|f(0.5, \dots, 0.5)\|_2^2$$

$$\begin{aligned} & + \left\| \frac{\partial f(x_1, 0.5, \dots, 0.5)}{\gamma_1 \partial x_1} \right\|_2^2 + \left\| \frac{\partial f(0.5, x_2, 0.5, \dots, 0.5)}{\gamma_2 \partial x_2} \right\|_2^2 + \dots \\ & + \left\| \frac{\partial f(x_1, x_2, 0.5, \dots, 0.5)}{\gamma_1 \gamma_2 \partial x_1 \partial x_2} \right\|_2^2 + \left\| \frac{\partial f(x_1, 0.5, x_3, 0.5, \dots, 0.5)}{\gamma_1 \gamma_3 \partial x_1 \partial x_3} \right\|_2^2 + \dots \\ & + \dots + \left\| \frac{\partial f(x_1, \dots, x_d)}{\gamma_1 \cdots \gamma_d \partial x_1 \cdots \partial x_d} \right\|_2^2 \end{aligned}$$



And now back to MCMC



Overcoming challenges of MCMC

- The probability densities where we use MCMC are typically **spiky** and often **multimodal**
 - Small step sizes may give samples that **do not represent** the distribution well
 - Large step sizes lead to **low acceptance** rates
- One may try **pilot** sampling or **multiple** chains
- **Parallel tempering** (replica exchange) combines benefits of large step sizes with multiple chains



Maximum likelihood estimation (MLE)

Let θ be a parameter to be estimated, and let

$q_{\text{DATA}}(\text{data}|\theta)$ = density for random DATA given parameter θ
= also known as **likelihood($\theta|\text{data}$)**

θ_{MLE} maximizes **likelihood(θ | data)** is called the **maximum likelihood estimate**

E.g. if the data is $y_0, \dots, y_{n-1} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then

$$q_{\text{DATA}}(\{y_i\}_{i=0}^{n-1} | (\mu, \sigma^2)) = \frac{1}{(2\pi\sigma)^{n/2}} \exp\left(-\sum_{i=0}^{n-1} (y_i - \mu)^2 / (2\sigma^2)\right)$$

$$\mu_{\text{MLE}} = \frac{1}{n} \sum_{i=0}^{n-1} y_i, \quad \sigma_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \mu_{\text{MLE}})^2$$



Bayesian Inference

$\varrho_{\text{prior}}(\theta) =$ density expressing **prior** belief about parameter Θ at θ

$\varrho_{\text{DATA}|\Theta}(\text{data}|\theta) =$ density for DATA given $\Theta = \theta$ evaluated at data
= also know as **likelihood** $_{\Theta|\text{DATA}}(\theta|\text{data})$

Then

$\varrho_{\text{posterior}}(\theta|\text{data}) =$ density expressing belief about Θ given data and prior belief

$$\begin{aligned} &= \frac{\varrho_{\text{joint}}(\theta, \text{data})}{\varrho_{\text{DATA}}(\text{data})} = \frac{\varrho_{\text{DATA}|\Theta}(\text{data}|\theta) \varrho_{\text{prior}}(\theta)}{\varrho_{\text{DATA}}(\text{data})} \\ &= \frac{\text{likelihood}_{\Theta|\text{DATA}}(\theta|\text{data}) \varrho_{\text{prior}}(\theta)}{\varrho_{\text{DATA}}(\text{data})} \propto \text{likelihood}_{\Theta|\text{DATA}}(\theta|\text{data}) \varrho_{\text{prior}}(\theta) \end{aligned}$$

Have **unnormalized** posterior, i.e. can ignore anything depending only on data

May compute maximum a posteriori (MAP) or posterior mean of θ through MCMC sampling



Queues

Imagine a customer service system with random arrival times of customers and random service times. There may be several stages. We consider the case where i represents an **event** (not a fixed time increment) E.g.,

A_i = arrival time of next customer $\stackrel{\text{IID}}{\sim} F_A$

N_i = number of customers in the queue

S_i = service time of first customer in queue $\stackrel{\text{IID}}{\sim} F_S$

Given $A[0] \sim F_A$, $N[0] \leftarrow 0$, $S[0] \sim F_S$, n #total number of customers

While $i < n$

 If $N[i] = 0$ #new customer

$N[i+1] \leftarrow 1$, $A[i+1] \sim F_A$, $S[i+1] \leftarrow S[i]$

 ElseIf $A[i] < S[i]$ #new customer

$N[i+1] \leftarrow N[i+1] + 1$, $A[i+1] \sim F_A$, $S[i+1] \leftarrow S[i] - A[i]$

 Else #finish serving customer

$N[i+1] \leftarrow N[i+1] - 1$, $A[i+1] = A[i] - S[i]$, $S[i+1] \sim F_S$

$i \leftarrow i+1$



Queues

Imagine a customer service system with random arrival times of customers and random service times. There may be several stages. Suppose that you want to measure customer waiting time and total time

A_i = arrival time of next customer $\stackrel{\text{IID}}{\sim} F_A$

N_i = number of customers in the queue

S_i = service time of first customer in queue $\stackrel{\text{IID}}{\sim} F_S$

T_i = total time passed

W_j = waiting time plus service time for the j^{th} customer



Queues

Given $A[0] \sim F_A$, $N[0] \leftarrow 0$, $S[0] \sim F_S$, $T[0] \leftarrow 0$

n #total number of customers

time #maximum time

$F \leftarrow 0$ #index of first customer in queue

$E \leftarrow 0$ #index of next customer to enter queue

While $i < n$ and $T[I] < \text{time}$

 If $N[i] = 0$ #get new customer

$N[i+1] \leftarrow 1$, $A[i+1] \sim F_A$, $S[i+1] \leftarrow S[i]$, $T[i+1] \leftarrow T[i+1] + A[i]$

$W[E] \leftarrow A[i]$, $E \leftarrow E+1$

 ElseIf $A[i] < S[i]$ #get new customer

$N[i+1] \leftarrow N[i+1] + 1$, $A[i+1] \sim F_A$, $S[i+1] \leftarrow S[i] - A[i]$, $T[i+1] \leftarrow T[i+1] + A[i]$

$W[E] \leftarrow A[i]$, $W[E-1] \leftarrow W[E-1] + A[i]$, ..., $W[F] \leftarrow W[F] + A[i]$, $E \leftarrow E+1$

 Else #finish serving customer

$N[i+1] \leftarrow N[i+1] - 1$, $A[i+1] = A[I] - S[i]$, $S[i+1] \sim F_S$, $T[i+1] \leftarrow T[i+1] + S[i]$

$W[E-1] \leftarrow W[E-1] + S[i]$, ..., $W[F] \leftarrow W[F] + S[i]$, $F \leftarrow F+1$

$i \leftarrow i+1$